Subhankar Biswas

DATA ENGINEER • AI / ML DEVELOPER • DATA & BI ANALYST • PYTHON SPECIALIST

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Enrolled M.Sc. Data Engineering Student: Versatile and results-driven M.Sc. Data Engineering candidate at Constructor University, Bremen, Germany with a bachelor's in physics, combining rigorous first-principles analytical modeling with production-grade software development. Proven expertise across end-to-end ETL/ELT pipeline design, distributed data architectures (Spark, Airflow, Kafka, DuckDB), cloud infrastructure (AWS, Docker, Terraform), applied AI/LLM engineering (RAG, PyTorch, Multi-Agent systems), and advanced business intelligence analytics (Power BI, Streamlit, SQL). Experienced in authoring resilient microservices, conducting causal inference, and collaborating with cross-functional technical teams to deliver high-impact, data-driven solutions.

CORE TECHNICAL COMPETENCIES & TOOLING

Languages & Scripting:	**Advanced:** Python (Pandas, NumPy, Scipy, Scikit-Learn), SQL • **Intermediate:** Bash/Shell • **Beginner:** R (Scripting & R Shiny), JavaScript/TypeScript
Data Engineering & Big Data:	**Advanced:** Distributed ETL/ELT, Data Modeling, Schema Normalization, Batch Pipelines • **Intermediate:** Apache Airflow, Apache Spark (PySpark), Streaming Pipelines
Databases & Storage:	**Advanced:** PostgreSQL, MySQL • **Intermediate:** DuckDB, MongoDB (NoSQL), Qdrant Vector DB, Time-Series Indexing, Data Warehousing • **Beginner:** Redis
AI, Deep Learning & LLMs:	**Advanced:** Large Language Models (LLMs), RAG Architectures, Multi-Agent Systems • **Intermediate:** PyTorch, Transformers, Computer Vision (CNNs), Causal AI
Cloud, DevOps & Tooling:	**Advanced:** Git/GitHub CI/CD, Linux/Unix, Docker, Docker Compose • **Intermediate:** FastAPI, Flask, AWS (EC2, S3, Lambda, API Gateway, CloudFormation)
BI, Analytics & Visualization:	**Advanced:** Microsoft Excel, Exploratory Data Analysis (EDA), Statistical Modeling • **Intermediate:** Streamlit, Plotly, Matplotlib, Seaborn, Power BI, A/B Testing • **Beginner:** Tableau

WORK & ACADEMIC EXPERIENCE

Cloud Solutions & Data Pipeline Architect (Projects)
04/2026 – 05/2026

Independent Engineering & Technical Integration • Germany

- Designed scalable serverless data pipelines on AWS utilizing Amazon S3, Lambda, API Gateway, and automated AWS CloudFormation templates.
- Containerized microservices using Docker and automated infrastructure provisioning, reducing operational deployment friction.
- Engineered automated data validation layers ensuring schema compliance and zero data loss during high-throughput ingestion runs.

Teaching Assistant – Computer Science & Data Systems
11/2018 – 08/2021

TechnoLab • Kalyani, West Bengal, India

- Mentored undergraduate and graduate students on database architectures (SQL/NoSQL), algorithmic problem solving, and Python programming.
- Conducted weekly tutorial sessions, evaluated practical lab assignments, and assisted students in debugging complex distributed data pipelines.
- Authored comprehensive lab documentation and coding standards to foster best practices in version control and test-driven development.

FEATURED ARTIFICIAL INTELLIGENCE & LLM PROJECTS

1. Semantic Document Intelligence & Enterprise RAG Platform

*Tech Stack**: Python, FastAPI, Qdrant Vector DB, OpenAI API, Neo4j, Apache Jena Fuseki, Celery, Docker, Semantic Chunking, RAG** • *GitHub Project Link*

- Engineered an enterprise-grade document intelligence platform processing unstructured multi-format documents (PDFs, technical papers, reports).
- Implemented semantic chunking, dense vector embeddings, and hybrid retrieval mechanisms in Qdrant, achieving sub-80ms semantic query latency.
- Built asynchronous REST API endpoints with FastAPI, incorporating metadata attribution, source verification, and containerized Docker deployment.

2. Autonomous Multi-Agent AI System & Decision Platform

*Tech Stack: **Python, Multi-Agent Protocols, Tool-Calling Orchestration, Streamlit, Agentic Workflows** • GitHub Project Link*

- Architected an autonomous multi-agent reasoning platform coordinating specialized LLM agents for dynamic task decomposition and execution.
- Implemented deterministic JSON schema validation and runtime verification middleware, eliminating agent hallucinations during tool calling.
- Shipped interactive Streamlit interfaces enabling real-time human-in-the-loop task inspection, telemetry monitoring, and decision tracking.

DATA ENGINEERING, BIG DATA & ETL PROJECTS

1. Production Binance Time-Series ETL & Airflow Pipeline

Tech Stack: ***Python, Apache Airflow, PostgreSQL, Docker, REST APIs, Data Quality Gates*** • *GitHub Project Link*

- Constructed an automated production ETL pipeline orchestrating daily extraction, transformation, and loading of high-frequency market streams.
- Configured DAG schedules in Apache Airflow with automated retry mechanisms, SLA alerting, and data validation gates in PostgreSQL.
- Optimized SQL indexing strategies and bulk ingestion pipelines, reducing database write latency and overall execution time by 35%.

2. Netflix Database Management & Streaming System

Tech Stack: ***Python, MongoDB (NoSQL), PostgreSQL (SQL), Docker Compose, Bash Scripting*** • *GitHub Project Link*

- Architected a polyglot database architecture benchmarked for relational structured transactions (PostgreSQL) and flexible JSON documents (MongoDB).
- Developed automated Python/Bash deployment scripts and Docker Compose environments for reproducible infrastructure provisioning.
- Implemented real-time data streaming simulators and aggregation pipelines evaluating throughput, indexing performance, and query efficiency.

DATA ANALYTICS, MACHINE LEARNING & CAUSAL AI PROJECTS

1. Bioprocess Digital Twin & Causal AI System

Tech Stack: ***Python, Causal Machine Learning, NumPy, SciPy, Time-Series Modeling, First-Principles Physics*** • *GitHub Project Link*

- Modeled non-linear dynamic biological trajectories from first-principles differential equations, simulating complex multi-phase states.
- Integrated causal inference to isolate latent biomarker interactions and quantify treatment effects under unobserved confounding variables.
- Formulated robust validation protocols, authoring comprehensive documentation and visualization dashboards for experimental analysis.

2. Delivery Operations & Funnel Analytics Dashboard

Tech Stack: ***Python, Streamlit, SQL, Geospatial Routing, Conversion Funnel Analytics, Plotly*** • *GitHub Project Link*

- Designed an interactive operational BI dashboard tracking order fulfillment KPIs, rider allocation efficiency, and customer conversion funnels.
- Applied spatial analysis and time-series aggregation to uncover supply-demand bottlenecks, identifying routing optimizations that cut transit delay.

3. Production CNN Classifier & GPU Optimization Pipeline

Tech Stack: ***Python, PyTorch, CUDA, Custom Training Loops, Regularization, Hyperparameter Optimization*** • *GitHub Project Link*

- Built and benchmarked deep convolutional neural network (CNN) architectures in PyTorch from scratch with custom forward/backward passes.
- Optimized CUDA memory allocation, mixed-precision training, and learning rate scheduling, reducing training epoch times by 30%.

EDUCATION

M.Sc. in Data Engineering | 2025 – Present
Constructor University – Bremen, Germany
Core Focus: Distributed Systems, Big Data Architectures, Machine Learning, Cloud Platforms, Python Programming, Data Pipelines.

PG Diploma in IT Project Management | 2016 – 2018
Welingkar Institute of Management – Mumbai, India

B.Sc. (Honours) in Physics | 2013 – 2016
West Bengal State University – India

CERTIFICATIONS & TRAINING

AI Tools & Engineering Certification | 2026
Be 10X • AI Tools & Claude Workshop **Certificate Link**

GenAI Hackathon Participant | 2026
Constructor University

LANGUAGES & SOFT SKILLS

- **English:** Full Professional / C1 (Fluent)
- **German:** Basic / A1 (Elementary)
- **Bengali:** Native | **Hindi:** Professional
- **Soft Skills:** Agile, Problem Solving, Cross-Functional Teamwork